Q1.i) Convert each binary number to decimal

a) 100001

$\overset{32}{1}\,\overset{16}{0}\,\overset{8}{0}\,\overset{4}{0}\,\overset{2}{0}\,\overset{1}{1}$

$32 + 1 = (33)_{10}$

b) $\overset{32}{1}\,\overset{16}{0}\,\overset{8}{0}\,\overset{4}{1}\,\overset{2}{1}\,\overset{1}{1}$

$32+4+2+1 = (39)_{10}$

c) $\overset{32}{1}\,\overset{16}{0}\,\overset{8}{1}\,\overset{4}{0}\,\overset{2}{1}\,\overset{1}{0}$

$32 + 8 + 2 = (42)_{10}$

d) $\overset{32}{1}\,\overset{16}{1}\,\overset{8}{1}\,\overset{4}{0}\,\overset{2}{0}\,\overset{1}{1}$

$32+16+8+1 = (57)_{10}$

e) $\overset{64}{1}\,\overset{32}{1}\,\overset{16}{0}\,\overset{8}{0}\,\overset{4}{0}\,\overset{2}{0}\,\overset{1}{0}$

$64+32 = (96)_{10}$

f) $\overset{128}{1}\,\overset{64}{1}\,\overset{32}{1}\,\overset{16}{1}\,\overset{8}{1}\,\overset{4}{1}\,\overset{2}{0}\,\overset{1}{1}$

$128+64+32+16+8+4+1$

$= (253)_{10}$

g) $\overset{128}{1}\,\overset{64}{1}\,\overset{32}{1}\,\overset{16}{1}\,\overset{8}{0}\,\overset{4}{0}\,\overset{2}{1}\,\overset{1}{0}$

$128+64+32+16+2$

$= (242)_{10}$

h) $\overset{128}{1}\,\overset{64}{1}\,\overset{32}{1}\,\overset{16}{1}\,\overset{8}{1}\,\overset{4}{1}\,\overset{2}{1}\,\overset{1}{1}$

$128+64+32+16+8+4$

$+ 2 + 1 = (255)_{10}$

Q1 (ii). What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits).

a) Two

$2^2 - 1 = 3$

b) Three

$2^3 - 1 = 7$

c) Four

$2^4 - 1 = 15$

d) Five

$2^5 - 1 = 31$

e) Six

$2^6 - 1 = 63$

f) Seven

$2^7 - 1 = 127$

g) Eight

$2^8 - 1 = 255$

h) Nine

$2^9 - 1 = 511$

i) Ten

$2^{10} - 1 = 1023$

Q1.(iii) List down all possible combinations for a 5-bit binary register.

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1
1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1

Q1 (iv). How many bits are required to represent the following decimal numbers.

a) 5
3 bits

c) 15
4 bits

e) 100
7 bits

g) 140
8 bits

b) 10
4 bits

d) 20
5 bits

f) 120
7 bits

h) 160
8 bits

Q2) i. Convert each hexadecimal number to binary and decimal

a) 46_{16}

4 6
0100 0110
$(01000110)_2$
$64+4+2 = (70)_{10}$

b) 54_{16}

5 4
0101 0100
$(01010100)_2$
$64+16+4 = (84)_{10}$

c) $B4_{16}$

B=11 4
1011 0100
$(10110100)_2$
$128+32+16+4 = (180)_{10}$

d) $1A3_{16}$

1 A=10 3
0001 1010 0011
$(000110100011)_2$
$256+128+32+2+1 = (419)_{10}$

e) FA_{16}

F=15 A=10
1111 1010
$(11111010)_2$
$128+64+32+16+8+2$
$=(250)_{10}$

f) ABC_{16}

A=10 B=11 C=12
1010 1011 1100
$(101010111100)_2$
$2048+512+128+32+16+8+4 = (2748)_{10}$

g) $ABCD_{16}$

A B C D=13
1010 1011 1100 1101
$(1010101111001101)_2$
$32768+8192+2048+512+256+128$
$+64+8+4+1 = (43981)_{10}$

Q2) ii) Convert each binary number to hexadecimal.

a) 1111

8 4 2 1 = 15

$(F)_{16}$

b) ~~11111~~

1011

8 4 2 1 = 11

B_{16}

c) 1 1111

0001 1111

1 8 4 2 1

$(1 \quad F)_{16}$

d) 1010 1010

8 4 2 1 = 10

$(AA)_{16}$

e) 1010 1100

8 4 2 1 8 4 2 1 = 12

$(AC)_{16}$

f) 1011 1011

8 4 2 1 = 11

$(B \quad B)_{16}$

Q2) iii. Convert each octal number to binary and decimal.

a) 14_8

1 4

421 421

001 100

$(001100)_2$

$8+4=(12)_{10}$

b) 53_8

5 3

421 421

101 011

$(101011)_2$

$32+8+2+1=(43)_{10}$

c) 67_8

6 7

421 421

110 111

$(110111)_2$

$32+16+4+2+1$

$=(55)_{10}$

d) 174_8

1 7 4

421 421 421

001 111 100

$(001111100)_2$

$64+32+16+8+4$

$=(124)_{10}$

e) 635_8

6 3 5

421 421 421

110 011 101

$(110011101)_2$

$256+128+16+8+4+1$

$=(413)_{10}$

f) 254_8

2 5 4

421 421 421

010 101 100

$(010101100)_2$

$128+32+8+4$

$=(172)_{10}$

g) 2673_8

2	6	7	3
4 2 1	4 2 1	4 2 1	4 2 1
010	110	111	011

$(0\,1\,0\,1\,1\,0\,1\,1\,1\,0\,1\,1)_2$

$1024+256+128+32+16+8$

$+2+1 = (1467)_{10}$

h) 7777_8

7	7	7	7
111	111	111	111

$(111\,111\,111\,111)_2$

12 bit binary register so max decimal value : $2^{12}-1$

$= (4095)_{10}$

Q2) iv. Convert each binary number to octal:

a) 100

$= 4_8$

d) 1 111

$(17)_8$

g) 110 011

6 3

$(63)_8$

b) 110

$4+2 = 6_8$

e) 11 001

$(31)_8$

h) 101 010

5 2

$(52)_8$

c) 1 100

$(14)_8$

f) 11 110

$(36)_8$

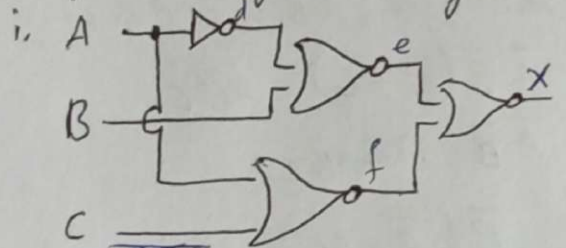
i) 10 101 111

2 5 7

$(257)_8$

Q3) Draw the truth table and verify the Boolean equation for the given Logic Diagrams.

i.



A	B	C	d	e	f	X
0	0	0	1	0	1	0
0	0	1	1	0	0	1
0	1	0	1	0	1	0
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	1	0	1	0	0
1	1	0	0	0	0	1
1	1	1	0	0	0	1

$X = \overline{\overline{\overline{A}+B} + \overline{A+C}}$

$X = (\overline{A}+B).(A+C)$

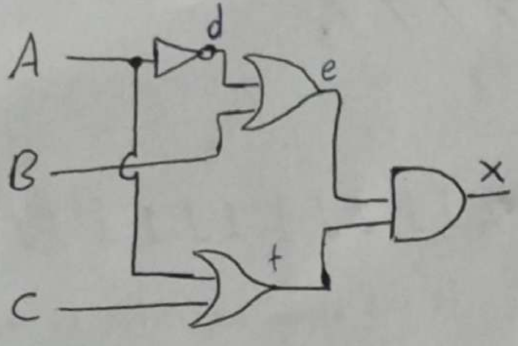
$X = (\overline{A}+B).(A+C)$

A \ BC	00	01	11	10
0		1	1	
1			1	1

$= \overline{A}C + BC + AB \rightarrow$ Redundant Literal

$= \overline{A}C + AB$

Q3) ii.



A	B	C	d	e	f	X
0	0	0	1	1	0	0
0	0	1	1	1	1	1
0	1	0	1	1	0	0
0	1	1	1	1	1	1
1	0	0	0	0	1	0
1	0	1	0	0	1	0
1	1	0	0	1	1	1
1	1	1	0	1	1	1

$(\bar{A}+B).(A+C)$

$\bar{A}A+\bar{A}C+AB+BC$

$0+\bar{A}C+AB+BC(A+\bar{A})$

$\bar{A}C+AB+ABC+\bar{A}BC$

$\bar{A}C+\bar{A}BC+AB+ABC$

$\bar{A}C(1+B)+AB(1+C)$

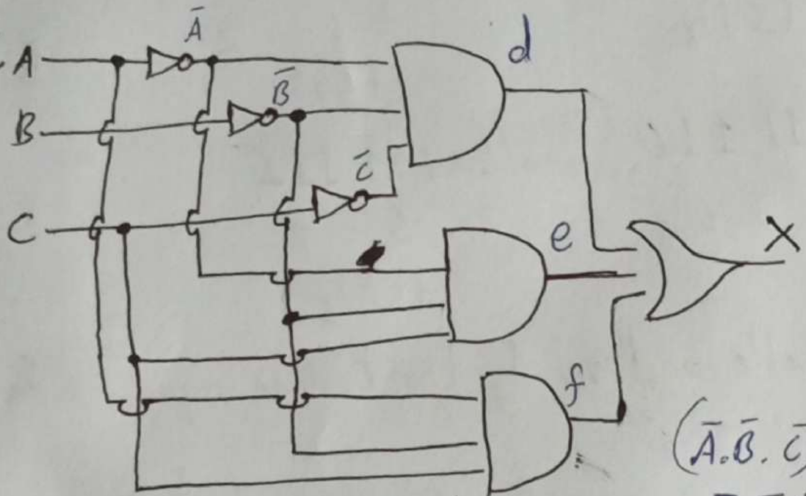
$\bar{A}C+AB$

A \ BC	00	01	11	10
0		1	1	
1			1	1

$\bar{A}C+BC+AB \rightarrow$ Redundant Literal Theorem

$\bar{A}C+AB$

Q3) iii.



A \ BC	00	01	11	10
0	1	1		
1		1		

$\bar{A}\bar{B}+\bar{B}C$

$A.\bar{B}.C$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	d	e	f	X
0	0	0	1	1	1	1	0	0	1
0	0	1	1	1	0	0	1	0	1
0	1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0	0
1	0	1	0	1	0	0	0	1	1
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	0	0	0

$(\bar{A}.\bar{B}.\bar{C})+(\bar{A}.\bar{B}.C)+(A.\bar{B}.C)$

$\bar{A}.\bar{B}(\bar{C}+C)+A\bar{B}C$

$\bar{A}.\bar{B}(1)+A\bar{B}C$

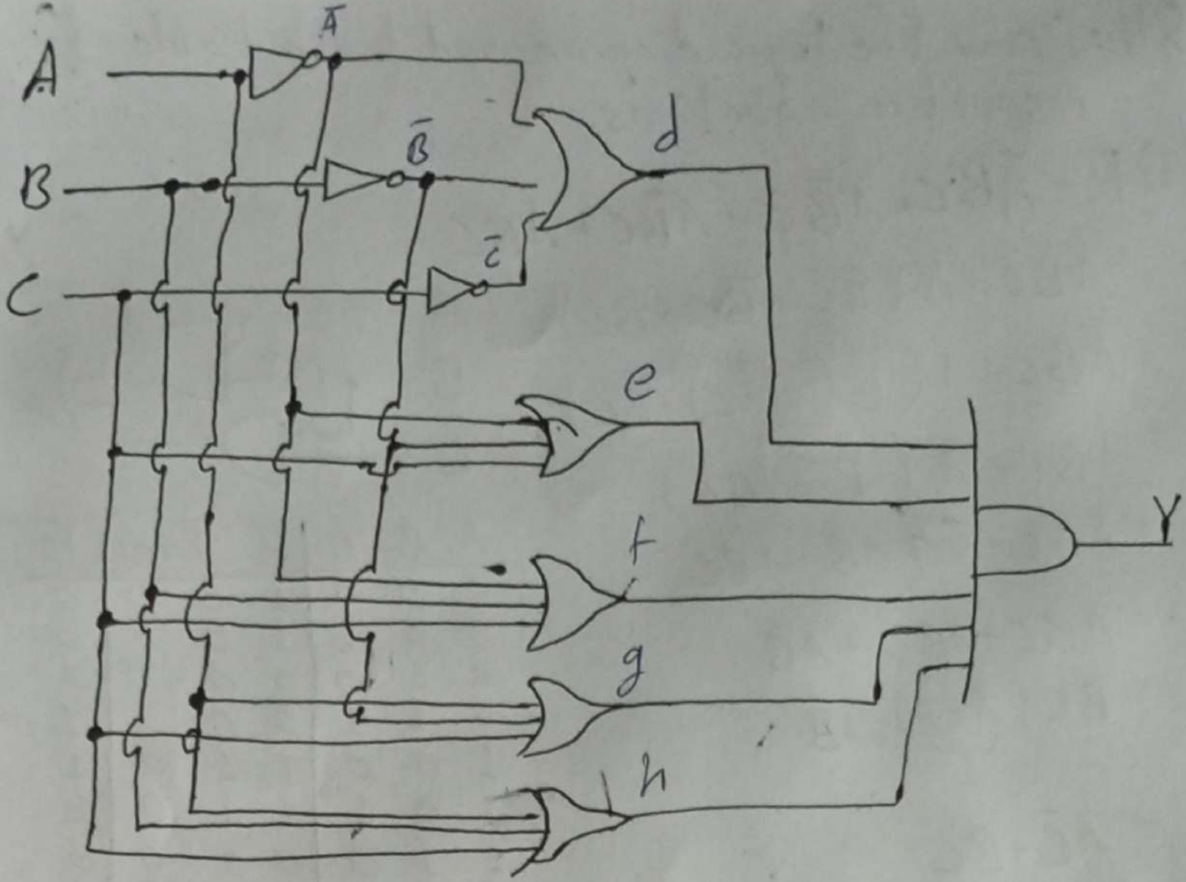
$\bar{B}(\bar{A}+AC)$

$\bar{B}((\bar{A}+A)(\bar{A}+C))$

$\bar{B}((1)(\bar{A}+C))$

$\bar{A}\bar{B}+\bar{B}C \,/\, \bar{B}(\bar{A}+C)$

Q3) iv.



A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	d	e	f	g	h	Y
0	0	0	1	1	1	1	1	1	1	0	0
0	0	1	1	1	0	1	1	1	1	1	1
0	1	0	1	0	1	1	1	1	0	1	0
0	1	1	1	0	0	1	1	1	1	1	1
1	0	0	0	1	1	1	1	0	1	1	0
1	0	1	0	1	0	1	1	1	1	1	1
1	1	0	0	0	1	1	0	1	1	1	0
1	1	1	0	0	0	0	1	1	1	1	0

A \ BC	00	01	11	10
0		1	1	
1		1		

$\bar{A}C+\bar{B}C$

Q4: Draw the logic diagram and truth table for the respective equations.

i) $F = \bar{A}B \cdot C + A\bar{B}\bar{C} + A\bar{B}C + ABC$

$= \bar{A}BC + A(\bar{B}\bar{C} + \bar{B}C + BC)$

$= \bar{A}BC + A(\bar{B}(\underbrace{\bar{C} + C}_{1}) + BC)$

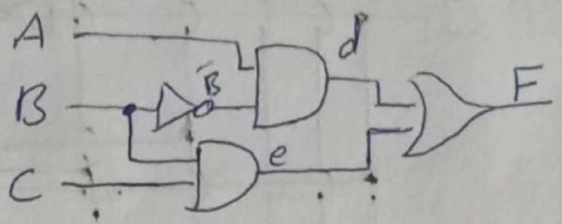
$= \bar{A}BC + A(\bar{B} + BC)$

$= \bar{A}BC + A\bar{B} + ABC$

$= \bar{A}BC + ABC + A\bar{B}$

$= BC(\underbrace{\bar{A} + A}_{1}) + A\bar{B}$

$= A\bar{B} + BC$

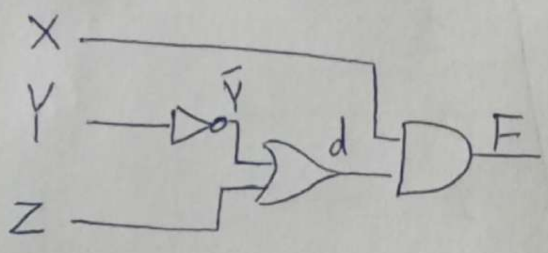


A	B	C	$\bar{B}$	d	e	F
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1

ii) $F = X\bar{Y} + XYZ$

$= X(\bar{Y} + YZ) \rightarrow$ Absorption Law

$= X(\bar{Y} + Z)$



X	Y	Z	$\bar{Y}$	d	F
0	0	0	1	1	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	1	1

CamScanner

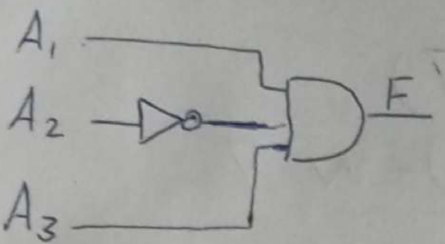
iii. $F = A_1\bar{A}_2A_3(A_1A_3 + A_1\bar{A}_2 \cdot A_3 + A_1A_2)$

~~=A~~ Opening Brackets and using identity $A \cdot A = A$.

$= A_1\bar{A}_2A_3 + A_1\bar{A}_2A_3 + A_1(\underbrace{\bar{A}_2 \cdot A_2}_{0})A_3$

$= A_1\bar{A}_2A_3 + A_1\bar{A}_2A_3 + 0 \qquad \because A + A = A$

$= A_1\bar{A}_2A_3$



A_1	A_2	A_3	$\bar{A}_2$	F
0	0	0	1	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	0
1	0	0	1	0
1	0	1	1	1
1	1	0	0	0
1	1	1	0	0

iv. $F = A\bar{B}\bar{C} + A\bar{B}C + AB(B + \bar{C})$

$= A\bar{B}\bar{C} + A\bar{B}C + AB + AB\bar{C}$

$= A(\bar{B}\bar{C} + \bar{B}C + B + B\bar{C})$

$= A(\bar{B}(\underbrace{\bar{C}+C}_{1}) + B(\underbrace{1+\bar{C}}_{1}))$

$= A(\underbrace{\bar{B} + B}_{1})$

$= A$

A —— F

A	F
0	0
1	1

v. $F = (A+B)(A' + AB) + (B' + AB')$

$= A\bar{A} + AB + \bar{A}B + AB + (B'(\underbrace{1+A}_{1}))$

$= 0 + AB + \bar{A}B + B'$

$= B(\underbrace{A + \bar{A}}_{1}) + B'$

$= B + B'$

$= 1$

vi. $WXY(W\bar{X}+W\bar{Y})+W\bar{X}Y(\bar{W}\bar{X}+\bar{X}\bar{Y}) = F$

$= W(\underbrace{X.\bar{X}}_{0})Y + WX(\underbrace{Y.\bar{Y}}_{0}) + (\underbrace{W.\bar{W}}_{0})\bar{X}Y + W\bar{X}(\underbrace{Y.\bar{Y}}_{0})$

$= 0 + 0 + 0 + 0$

$= 0$

vii. $F = (WX + W\bar{Y})(W+X) + WX(\bar{X}' + \bar{Y}')$

$= WX + WX + W\bar{Y} + WX\bar{Y} \underbrace{+ W(\underbrace{X.\bar{X}}_{0}) + WX\bar{Y}}_{+0}$

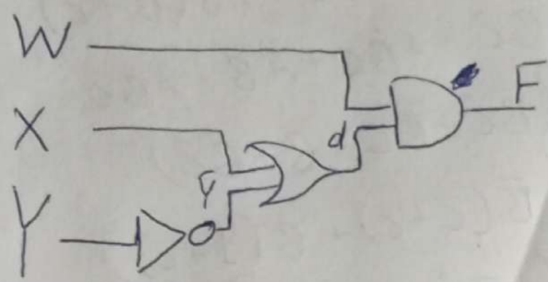
$= WX + W\bar{Y} + WX\bar{Y} \quad \cancel{+WXY}$

$= W(X + \bar{Y} + X\bar{Y})$

$= W(X + \bar{Y}(\underbrace{1+X}_{1}))$

$= W(X+\bar{Y})$

W	X	Y	$\bar{Y}$	d	F
0	0	0	1	1	0
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1



viii. $F = (EP + E\bar{N})(\bar{E}P + E\bar{N}' + \bar{N}P) + (\bar{E}P + E\bar{N}')(E\bar{N}' + \bar{N}P)$

$= (E.\bar{E})P + E\bar{N}P + E\bar{N}P + (E.\bar{E})\bar{N}P + E\bar{N} + E\bar{N}P + (E.\bar{E})\bar{N}P + \bar{E}\bar{N}P + \bar{E}\bar{N} + E\bar{N}P$

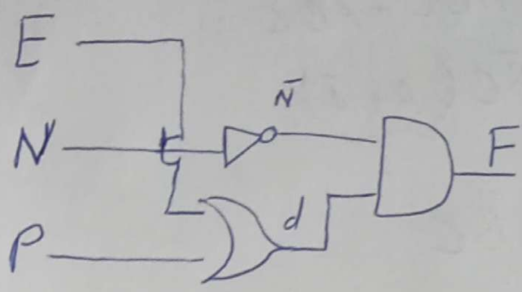
$= 0 + E\bar{N}P + 0 + E\bar{N} + 0 + \bar{E}\bar{N}P$

$= E\bar{N}P + \bar{E}\bar{N}P + E\bar{N}$

$= \bar{N}P\underbrace{(E+\bar{E})}_{1} + E\bar{N}$

$= \bar{N}P + E\bar{N}$

$= \bar{N}(P+E)$



E	N	P	$\bar{N}$	d	F
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	0	1	0

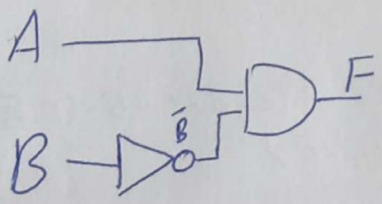
ix. $F = A\bar{B} + A(\overline{B+C}) + B(\overline{B+C})$

$= A\bar{B} + A(\bar{B}.\bar{C}) + \underbrace{B(\bar{B}.\bar{C})}_{0}$

$= A\bar{B} + A\bar{B}\bar{C}$

$= A\bar{B}(1+\bar{C})$

$= AB'$



A	B	$\bar{B}$	F
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

x. $F = [A\bar{B}(C+BD) + \bar{A}\bar{B}]C$

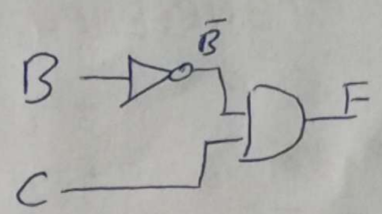
$= C(A\bar{B}C + A\underbrace{(B.\bar{B})}_{0}D + \bar{A}\bar{B})$

$= C(A\bar{B}C + \bar{A}\bar{B})$

$= A\bar{B}C + \bar{A}\bar{B}C$

$= \bar{B}C\underbrace{(A+\bar{A})}_{1}$

$= \bar{B}C$



B	C	$\bar{B}$	F
0	0	1	0
0	1	1	1
1	0	0	0
1	1	0	0

Q5: Draw K-Map and reduce the equation to verify.

i. $F(A,B,C) = \Sigma(0,2,3,7) = \bar{A}\bar{C} + CD$

A \ BC	00 (0)	01 (1)	11 (3)	10 (2)
0 (0)	1	0	1	1
1 (4)	0	0	1	0

$\boxed{BC + \bar{A}B + \bar{A}\bar{C}}$

The equation in question is invalid

ii. $F(A,B,C) = \Sigma(0,2,4,5,6,7) = A + C'$

A \ BC	00 (0)	01 (1)	11 (3)	10 (2)
0 (0)	1	0	0	1
1 (4)	1	1	1	1

$\boxed{A + \bar{C}}$

iii. $F(A,B,C) = \Pi(0,1,5,6) = A'B + BC + AB'C'$

A \ BC	00 (0)	01 (1)	11 (3)	10 (2)
0 (0)	0	0		
1 (4)		0		0

$\cancel{AB + B\bar{C} + \bar{A}B\bar{C}}$

$= (A+B)(B+\bar{C})(\bar{A}+\bar{B}+C)$

$= (AB + A\bar{C} + B + B\bar{C})(\bar{A} + \bar{B} + C)$

$= (A.\bar{A})B + A(B.\bar{B}) + ABC + (A.\bar{A})\bar{C} + A\bar{B}\bar{C} + A(C.\bar{C}) + \bar{A}B + (B.\bar{B})$
$+ BC + \bar{A}B\bar{C} + (B.\bar{B})\bar{C} + B(C.\bar{C})$

$= ABC + A\bar{B}\bar{C} + \bar{A}B + BC + \bar{A}B\bar{C}$

$\cancel{A(BC + \bar{B}\bar{C})} = A\bar{B}\bar{C} + BC + ABC + \bar{A}B + \bar{A}B\bar{C}$

$= A\bar{B}\bar{C} + BC + ABC + \bar{A}B(1 + \bar{C})$

$= A\bar{B}\bar{C} + BC + B(AC + \bar{A})$ ← Absorption Law

$= A\bar{B}\bar{C} + BC + B(\bar{A} + C)$

$= A\bar{B}\bar{C} + \underbrace{BC + \bar{A}B + BC}$

$\boxed{= A\bar{B}\bar{C} + BC + \bar{A}B}$

iv. $F(A,B,C) = AC + \bar{B}\bar{C} + AB\bar{C} = A + \bar{B}\bar{C}$

$\cancel{= AC + AB\bar{C} + \bar{B}\bar{C}}$

$\cancel{= A(C + B\bar{C}) + \bar{B}\bar{C}}$

$= A\_C + \_\bar{B}\bar{C} + AB\bar{C}$

5: 1 0 1   0: 0 0 0   1 1 0 :6

7: 1 1 1   4: 1 0 0

A \ BC	00 (0)	01 (1)	11 (3)	10 (2)
0 (0)	1			
1 (4)	1	1	1	1

$\boxed{A + \bar{B}\bar{C}}$

$F(A,B,C) = \Sigma(0,4,5,6,7)$

V. $F(A,B,C,D) = \Sigma(0,2,4,7) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{D} + \bar{A}BCD$

AB \ CD	00 (0)	01 (1)	11 (3)	10 (2)
00 (0)	1			1
01 (4)	1		1	
11 (12)				
10 (8)				

$\boxed{\bar{A}\bar{B}\bar{D} + \bar{A}\bar{C}\bar{D} + \bar{A}BCD}$

The equation in question is invalid.

Vi. $F(A,B,C,D) = \Sigma(3,7,11,15) = CD$

AB \ CD	00 (0)	01 (1)	11 (3)	10 (2)
00 (0)			1	
01 (4)			1	
11 (12)			1	
10 (8)			1	

$\boxed{CD}$

Vii. $F(A,B,C,D) = \Sigma(1,3,5,7,11,15) = CD + A'D'$

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11			1	
10			1	

$\boxed{CD + A'D}$

The equation in question is invalid.

viii. $F(A,B,C,D) = \sum(1,3,5,7,10,11,14,15) = A'D + AC$

	AB \ CD	00 (0)	01 (1)	11 (3)	10 (2)
0	00		1	1	
4	01		1	1	
12	11			1	1
8	10			1	1

$\boxed{\bar{A}D + AC}$

ix. $F(A,B,C,D) = \sum(0,2,4,5,7,10) = \bar{A}\bar{C}\bar{D} + \bar{A}BD + \bar{B}C\bar{D}$

	AB \ CD	00 (0)	01 (1)	11 (3)	10 (2)
0	00	1			1
4	01	1	1	1	
12	11				
8	10				1

$\boxed{\bar{A}\bar{C}\bar{D} + \bar{A}BD + \bar{B}C\bar{D}}$

x. $F(A,B,C,D,E) = \sum(0,1,2,3,4,5,6,7,10,19,23,25,27,29,31) = \bar{A}\bar{B} + \bar{A}\bar{C}D\bar{E} + ABE + ADE/\bar{B}D$

	AB \ CDE	000 (0)	001 (1)	011 (3)	010 (2)	110 (6)	111 (7)	101 (5)	100 (4)
0	00	1	1	1	1	1	1	1	1
8	01				1				
24	11		1	1			1	1	
16	10			1			1		

$\bar{A}\bar{B} + \bar{A}\bar{C}D\bar{E} + AB\bar{C}E + ABCE + \bar{B}\bar{C}DE + \bar{B}CDE$

$\bar{A}\bar{B} + \bar{A}\bar{C}D\bar{E} + ABE\underbrace{(C+\bar{C})}_{1} + \bar{B}DE\underbrace{(\bar{C}+C)}_{1}$

$\boxed{\bar{A}\bar{B} + \bar{A}\bar{C}D\bar{E} + ABE + \bar{B}DE}$

CamScanner